

A MULTIFUNCTIONAL ROLE FOR OCTOPAMINE IN LOCUST FLIGHT

Ian Orchard

Department of Zoology, University of Toronto, Toronto, Ontario Canada M5S 1A1

Jan-Marino Ramirez

Fachbereich Biologie, Universität Kaiserslautern, W-6750 Kaiserslautern, Germany

Angela B. Lange

Department of Zoology, Erindale College, University of Toronto, Mississauga, Ontario, Canada L5L 1C6

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PERSPECTIVES AND OVERVIEW

A complex cascade of neuronal, hormonal, muscular, and metabolic events leads to the initiation and maintenance of a behavioral act. Hormones are released to mobilize metabolic energy; muscle properties are altered to ensure the generation of the necessary force; neuronal circuits are activated to generate a specific locomotor pattern; and the neuronal circuit underlying respiration may also be altered to adapt the animal to increased oxygen consumption. All these events are part of a complex transition of the animal into a new behavioral state. A critical aspect of this transition is the need to alter these events at the appropriate time. This coordinating function may not only depend on neuronal interaction but may also be attributed to chemical substances that bias, at many levels, neuronal, hormonal, and muscular events towards the new functional state of the animal. For example, egg-laying by *Aplysia* spp. requires several peptides (38, 76) to elicit a complex of behaviors, including the cessation of locomotion, increase in heart and respiratory rate, egg laying, head waving, and mucous secretion. Similarly, serotonin controls

the feeding behavior of several species, including the leech (see 37), several molluscan species (see 14, 34), and the insect *Rhodnius prolixus* (see 3). However, because the effects, often at numerous levels, that result in a behavioral act are so complex, we have only a preliminary understanding of the coordinating role of chemical substances. Nevertheless, a few reviews have already addressed this issue, and the reader is directed to these for a fuller appreciation of the integrating actions of chemical substances on behavior (6, 27, 33, 34, 37). Here, we address this issue for the flight system of the locust, a locomotor system that has drawn much attention over the past decade. Much is known about the neuronal mechanisms involved in the generation and initiation of flight, the hormones released during flight, and the metabolic and respiratory processes that occur during flight. Recent studies on this system indicate that the release of the biogenic amine, octopamine, may play a major role in the coordination of these events. We describe the physiological processes at the onset of flight and emphasize the role of octopamine in adapting the animal to the altered conditions occurring during the first minutes of flight. As we shall see, octopamine not only promotes the expression of the flight motor output in locusts, but it also integrates flight performance at several levels of organization; it biases both nervous and nonnervous tissues towards a more efficient adaptive behavior, acting as neurotransmitter, neuromodulator, and neurohormone. In the subsequent sections we consider: (a) the role of octopamine in regulating energy metabolism at the onset of flight, (b) the role of octopamine in setting up the neuronal processes at the onset of flight, (c) octopaminergic modulation of flight muscles, (d) the role of endogenously released octopamine in the control of locust flight, and (e) octopaminergic modulation of insect behavior.

By discussing each of these aspects, we hope to illustrate that octopamine acts in a coordinated manner to prepare the locust for this very energy-demanding behavior. However, we emphasize in each section the limitations in interpreting the functional significance of the octopaminergic modulation for the flight performance. One should also be aware that the octopaminergic modulation of locust flight can only represent a small part of a complex scenario in which many modulatory substances affect several aspects of different behaviors, which all together lead to a "normal-behaving animal," a scenario that we are only just beginning to understand.

THE ROLE OF OCTOPAMINE IN REGULATING ENERGY METABOLISM AT THE ONSET OF FLIGHT

Insect flight is an extremely energy-demanding behavior in which metabolic activity increases 50- to 100-fold in the transition from rest to flight (4). Because the flight muscles store only small amounts of energy-rich substrates,

these substrates must be mobilized from storage sites (generally the fat body), transported via the hemolymph, and unloaded into the flight muscles, which may also be biased towards the appropriate substrate oxidation.

During the first 20–30 min of flight, hemolymph trehalose is the predominant energy source in locusts. However, the concentration of hemolymph trehalose drops after 30 min of flight to a steady state of 35% of the resting level. The steady state is the result of utilization and mobilization mainly from fat body glycogen stores (4), during which trehalose provides only 25% of the total energy supply. At this time, flight activity is accompanied by a marked elevation in hemolymph lipid (diacylglycerol) levels, and lipid provides 75% of the total energy supply. These elevated lipid levels, characteristic of locust flight, are hormonally controlled, with octopamine playing a dual role in this process (42, 48). Octopamine acts as a neurotransmitter controlling the release of peptidergic adipokinetic hormones, as well as a neurohormone with direct adipokinetic activity on fat body. In addition, octopamine stimulates oxidative metabolism in the flight muscles (9).

Octopamine and the Release of Peptidergic Adipokinetic Hormones

Neurosecretory cells in the glandular lobe of the corpus cardiacum of locusts are the source of AKH I and AKH II, two peptidergic adipokinetic hormones (AKHs), which are released in response to flight (42, 45, 46). These hormones mediate the mobilization of lipid from the fat body and may also participate in trehalose mobilization (4, 42). Thus, the AKHs provide energy sources for sustained flight. Electrical stimulation of the nervi corporis cardiaci II (NCC II), which project into the glandular lobe from the brain, results in an elevation in cyclic AMP (cAMP) content in the glandular lobe (44, 49) and in the release of both AKHs (42, 45). The neurally evoked release of AKHs involves synapses between axons of NCC II and the neurosecretory cells, and the increase in cAMP may be coupled to the process of synaptic transmission (44, 52, 53). These well-described (59, 61) synapses are characterized by the presence of large electron-dense granules (100 nm diameter) and numerous small clear vesicles. Anatomical and physiological evidence suggests that octopamine is the natural transmitter of these synapses. Octopamine is present within the glandular lobe and NCC II (51), and the ultrastructural features of the vesicles are similar to those of identified octopaminergic neurons (30). Two experiments indicate the aminergic nature of the transmitter. Elevation of cAMP and neurally evoked release of AKHs are abolished (*a*) after depletion of aminergic stores by pretreatment with reserpine and (*b*) by treatment with α -adrenergic receptor antagonists (44, 49, 52, 53). That this aminergic transmitter is octopamine is suggested by the finding that exogenous application of low concentrations of octopamine (10^{-7} M) to the in vitro

glandular lobe mimics the biochemical and physiological effects of nerve stimulation. The responses to octopamine are dose dependent with half-maximal stimulation at 5×10^{-6} M and maximal stimulation at 10^{-4} M. These responses are again blocked by α -adrenergic receptor antagonists. The ability of several putative aminergic agonists and antagonists to mimic or abolish the neurophysiological responses have been used to classify the receptors. The receptors mediating the elevation in cAMP and release of AKHs exhibit a specificity for the monophenolic amines octopamine and synephrine; the receptors appear to belong to the octopamine-2 class (52). Cyclic AMP plays an integral role in the release of the AKHs (53). The phosphodiesterase inhibitor, IBMX, and the diterpene, forskolin, which is an activator of adenylate cyclase, both mediate the release of AKH I and AKH II from in vitro corpora cardiaca. The cAMP analogs dibutyryl cAMP and 8-bromo cAMP also elicit release of AKHs, and IBMX potentiates octopamine-stimulated release of hormones. Thus, these data indicate that cAMP participates in regulating hormone release from the neurosecretory cells of the glandular lobe and support the hypothesis that octopamine, released synaptically, mediates release, at least in part, via changes in intracellular levels of cAMP. The results of the above experiments are all consistent with octopamine being the natural transmitter within this neurosecretory system.

Octopamine as a Neurohormone with Adipokinetic Activity

Octopamine also acts as a neurohormone in locusts. When tested on locust fat body in vivo and in vitro, octopamine has adipokinetic activity (43, 84, 86). This observation, coupled with the demonstration that hemolymph octopamine titers increase dramatically during the first few minutes of flight (24, 25, 42), has led to the suggestion that octopamine is itself an adipokinetic hormone (see 48). With this in mind, one could ask, how does octopamine interact with the peptidergic AKH I and AKH II during flight? Examination of the increase in hemolymph lipid concentration during flight in *Locusta migratoria* reveals a biphasic pattern. Initially, levels increase within the first 10 min of flight, and a second, larger increase begins approximately 10 min later (46). Clearly, the control of hemolymph lipid during flight is not simple and involves the integration of many hormones. Three of these hormones, octopamine, AKH I, and AKH II, have been identified and their titers during flight measured (24, 46). A model incorporating the events known to occur during flight has been proposed (see 42, 48) and is repeated here.

As can be seen in the model (Figure 1), the flight-induced increases in these hormones occur with different time courses. A short-lived elevation in hemolymph octopamine peaks at 10–15 min of flight, fat body cAMP levels increase concomitantly. Because locust fat body responds in vitro to octopamine with an increase in cAMP and release of lipid (43, 86), we attribute the

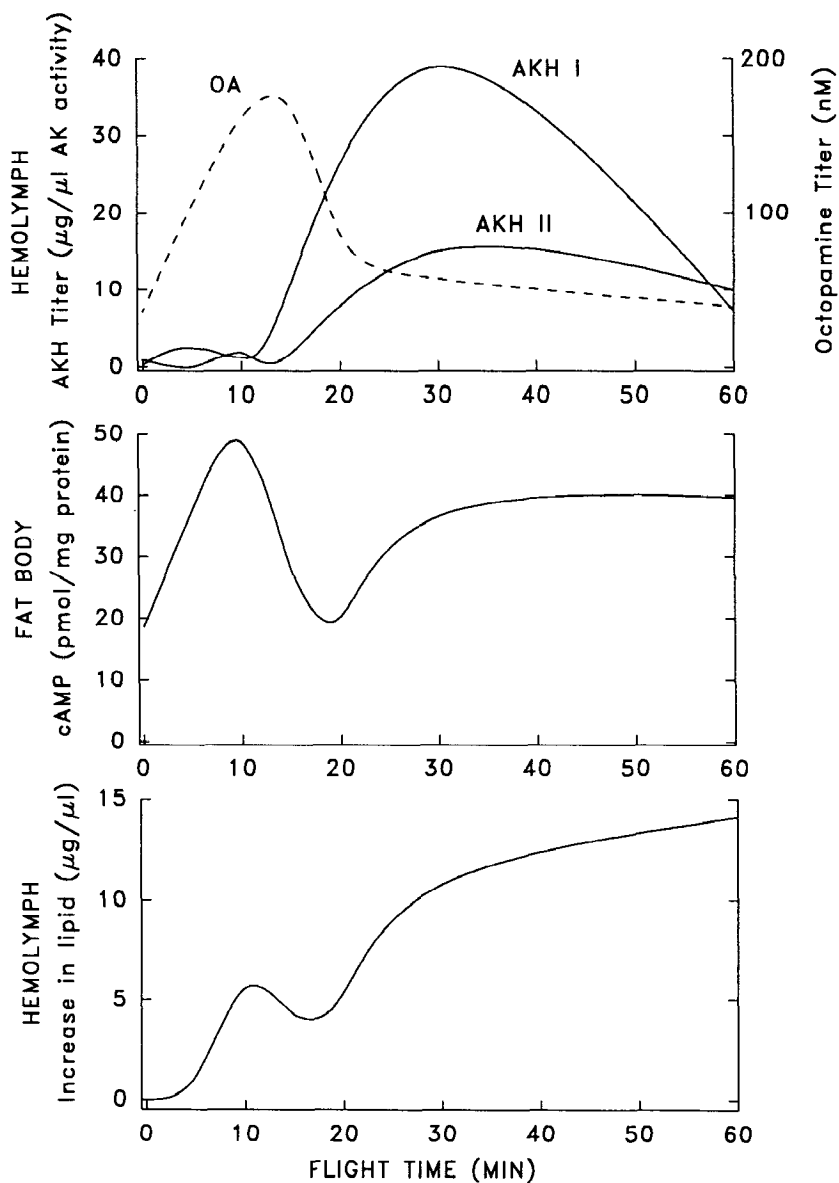


Figure 1 Metabolic hormonal events associated with flight in locusts. For complete explanation see text. Octopamine titer redrawn from Ref. 24. AKH I and II titers and increase in hemolymph lipid redrawn from Ref. 46. Fat body cAMP levels redrawn from Ref. 47. Model is modified from Ref. 48.

initial rise in hemolymph lipid during flight to the presence of hemolymph octopamine.

Turning our attention to the peptides that are involved with lipid regulation, one can see that the titer of AKH I begins to elevate after approximately 15 min of flight, while the titer of AKH II elevates between 15–30 min of flight. The appearance of these two AKHs coincides with the second phase of cAMP elevation, which occurs between 20–30 min of flight. Several groups have suggested that the peptidergic control of lipid release in locusts is also mediated, at least in part, via cAMP (43, 47, 79, 86). Thus, the second phase of cAMP elevation probably results in the second rise in hemolymph lipid, and can also be attributed to the hemolymph AKHs.

Experimental interference with the release of these hormones (47) has supported the contention that the changes in fat body cAMP reflect the presence of octopamine and AKHs. The second increase in cAMP (measured at 30 min), but not the first increase (measured at 10 min), is eliminated by neck ligation, a procedure that prevents the AKHs from reaching the rest of the animal but does not prevent the increase in octopamine titer. Injection of 6 mg trehalose delays the release of AKHs during flight (11) and inhibits flight-induced release of octopamine (47). Trehalose injection also eliminates the increases in fat body cAMP normally induced by flight, thereby demonstrating again a correlation between octopamine and AKHs in this cAMP elevation.

In summary, we believe that the rise and fall in octopamine titer results in an elevation and decline in fat body cAMP that stimulates the initial mobilization of lipid seen within the first 10 min of flight. The AKHs are released after approximately 15 min, following activation of octopaminergic synapses in the glandular lobe, and the AKHs elevate fat body cAMP, thereby stimulating a second mobilization of lipid. In this way, energy-rich substrates are mobilized for transport to the flight muscles to help drive the energy-demanding activity of flight.

THE ROLE OF OCTOPAMINE IN SETTING UP THE NEURONAL PROCESSES AT THE ONSET OF FLIGHT

In addition to metabolic adaptations, flight also requires adaptations in oxygen supply. Oxygen consumption increases 24-fold at the onset of flight and major alterations in the respiratory system are necessary to provide sufficient oxygen. The respiratory rate in the abdomen increases, and elements of the rhythm-generating neural network located in the subesophageal ganglion change their rhythmicity and discharge (68, 69). These alterations coincide with the initiation of flight and in anticipation of the increased oxygen consumption. A feed-forward mechanism is controlled by the same neurons that are involved

in the initiation of the flight motor pattern (62, 69). The possibility that octopamine affects the respiratory system at the onset of locust flight has not been examined in detail. However, it has been demonstrated that octopamine can increase the excitability of certain respiratory neurons (71), and for another insect species, *Corydalis cornutus*, studies have shown that octopamine can increase respiratory rate (5). An increase in respiratory rate following octopamine injection has also been described for the locust (77, 78). However, this increase occurs simultaneously with the initiation of flight and therefore does not necessarily indicate a direct effect on the respiratory system.

More is known about the effects of octopamine on flight rhythm generation, the most obvious neuronal process occurring at the onset of flight. Octopamine, injected into certain neuropile regions of thoracic ganglia, can induce flight-rhythmic activity in locusts and the moth, *Manduca sexta* (12, 77, 78, 80, 81). The induction of such rhythmic activity results at least in part from octopamine acting within the central nervous system (CNS) because this activity can be induced by the injection of octopamine into the isolated metathoracic ganglion of the locust (81). Octopamine also affects sensory pathways. Following octopamine perfusion, proprioceptive pathways from the wing are enhanced (12, 72). The cellular mechanisms underlying these effects and the functional relevance of octopaminergic modulation for the generation of flight have been investigated in the locust (65, 71–73). As demonstrated in the following sections, octopamine, acting centrally, induces bursting properties in various flight interneurons and increases, peripherally, the responsiveness of wing proprioceptors.

The Central Effects of Octopamine on Flight Interneurons

An important prerequisite in understanding the cellular mechanisms underlying the actions of octopamine is a detailed knowledge of the neuronal mechanisms leading to the generation of the flight motor pattern. The latter is generated by the interaction of an interneuronal network located in the thoracic and fused abdominal ganglia as well as in the wing proprioceptors located at the base of each wing (90). Many interneurons within this network have been identified, and their activity in deafferented and intact flying locusts has been determined. Also, much is known about the connectivity between interneurons and wing proprioceptors (55, 57, 75, 89). Recent investigations indicate that the generation of the flight rhythm relies not only on the characteristics of synaptic connectivity (network properties) but also on the ability of flight interneurons to generate plateau potentials (cellular properties) (73). These potentials are prolonged regenerative depolarizations that promote bursts of activity. However, the expression of these bursting properties is conditional: these properties are absent in quiescent animals and present during flight-rhythmic activity. The induction of bursting properties during flight has

been observed for several important elements of the neuronal circuit for flight (72), leading to the conclusion that this circuit is transformed at the onset of flight from a passive into a dynamic state. Several lines of evidence suggest that octopamine might be involved in the induction of bursting properties at the onset of flight. Measurements of the octopamine content in the metathoracic ganglia indicate that the octopamine concentration increases significantly during the first 20 min of flight (16). This increase in octopamine content could result from the dorsal unpaired median (DUM) neurons (see 41), which are octopaminergic neurons that are activated at the onset of flight (65). These neurons have output synapses within the dorsal neuropile of the thoracic ganglion (87), a neuropile region that contains most flight interneurons (67). Injection of octopamine into this region can induce flight-rhythmic activity (77, 81, 82). This finding may be explained by the ability of octopamine to induce burst-promoting membrane properties in flight interneurons (71, 72). Following octopamine application, depolarizing current pulses evoke rhythmic bursting in various flight interneurons. Figure 2 shows the induction of bursting properties for interneuron 566. A current pulse injected into interneuron 566 of a quiescent locust evokes only tonic activity in the absence of octopamine, while rhythmic bursting is evoked by the same current pulse following octopamine application. The bursting is similar to that seen in interneuron 566 during flight. The frequency of bursting is close to the frequency generated by the central rhythm generator (~ 10 Hz). The injection of hyperpolarizing currents indicates that this bursting is generated intrinsically. Hyperpolarizing current pulses suppress the generation of a burst and thus reset the bursting rhythm (72). Hyperpolarizing offset currents lead to sudden nonlinear changes in the depolarization amplitude during octopamine perfusion (71, 72), which are also seen when the offset currents are gradually changed during flight. Following octopamine application, bursts of activity similar to those observed during flight can also be triggered by synaptic input from wing proprioceptors (73). This finding could explain how octopamine enhances proprioceptive pathways (12). Consistent with the notion that bursting properties are induced by a chemical substance released during flight is the observation that burst generation is often increased directly after flight (63).

The induction of plateau and bursting properties by neuromodulators as described for octopamine is a general phenomenon. It has been found in the crustacean stomatogastric system (see 32), in certain molluscan neurons (2), in cat and turtle spinal motoneurons (see 13, 29), in guinea pig respiratory neurons (17), and in lamprey spinal neurons (85). The transformation of a neuronal circuit from a passive into a dynamic state by neuromodulators might also be a general phenomenon for many motor systems. For example, the neuronal circuit for walking in cats cannot be activated in animals in which

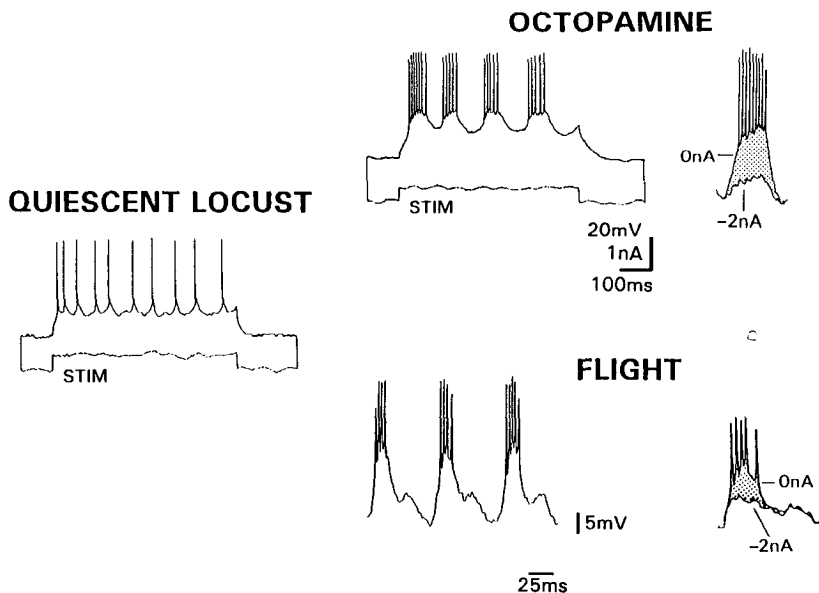


Figure 2 Induction of bursting properties in interneuron 566. In a quiescent animal, a depolarizing current pulse (left panel, lower trace) evokes tonic activity in interneuron 566 (left panel, upper trace). Following application of octopamine (upper middle panel), a similar pulse evokes rhythmic bursting (from 72). Injection of hyperpolarizing DC offset (-2 nA) causes a sudden drop in amplitude of an octopamine-induced burst in interneuron 566 (upper right panel) as indicated by two superimposed recordings obtained at 0 nA and -2 nA DC offset (J.-M. Ramirez, unpublished observation). In intact tethered flying locusts, interneuron 566 is rhythmically active in phase with the flight rhythm (lower middle panel). Nonlinear amplitude drops like those seen for the octopamine-induced burst are also found for the rhythmic bursts generated in interneuron 566 during flight (lower right panel, two recordings superimposed).

descending pathways have been acutely severed by spinalization. However, the neuronal circuit for walking in these acute spinal cats can be activated by the application of clonidine and L-DOPA (22, 26). A similar case exists for the neuronal circuits of the crustacean stomatogastric system. These circuits are mostly inactive when descending pathways to this system are transected but can be activated by application of amines or peptides (27).

The Peripheral Effects of Octopamine on Wing Proprioceptors

Octopamine not only acts upon central neurons but also upon neurons in the periphery, resulting in a direct modulation of proprioceptive input from wing proprioceptors. As mentioned above, the generation of the flight motor pattern strongly depends on phasic input from wing proprioceptors. The two most important proprioceptors for flight are: the hindwing tegula and the forewing

stretch receptor (SR), a proprioceptor that is considered in the following section in more detail. The SR is a single-celled proprioceptor located at the base of each wing from where it projects into the thoracic ganglia (1) and connects onto various flight neurons (90). In the intact flying locust, the SR is rhythmically active at the beginning of each wing depression (55), and stimulation of the SR can reset and entrain the flight rhythm (56, 74). The functional significance of the stretch receptor may be demonstrated by removing the influence of this sensory neuron, a procedure that causes a 60% drop in wingbeat frequency (70).

As with neurons in the CNS, octopamine seems to set the activity of the SR to a dynamic state. Injection of octopamine into the hemolymph enhances the responsiveness of the SR to sinusoidal wing movements (65). For example, a SR that responds with one spike per wing movement discharges occasionally with two spikes after application of octopamine at a concentration of 5×10^{-7} M, and the SR consistently discharges with two spikes at a concentration of 5×10^{-5} M (Figure 3). Not only the number of spikes but also the timing of the SR activity is affected by octopamine injection. The interval between the two spikes decreases with increasing octopamine concentration (Figure 3). During the first minutes of flight, hemolymph octopamine levels rise to 2×10^{-7} M (24, 42). This concentration is within the dose-response range over which octopamine modulates the activity of the SR (Figure 3). That blood-borne octopamine has access to the SR has been demonstrated by injecting octopamine into the hemolymph of intact locusts. The activity in the SR increases usually 9–12 s after injection of octopamine (Figure 3).

A typical feature of the SR modulation is that the SR activity becomes saturated if octopamine is injected over short intervals. In such cases, the SR responds consistently after reaching a peak at a relatively high discharge rate. Further injection of octopamine is ineffective (Figure 3). Of functional interest is the observation that the SR response has similar saturation characteristics directly after dissection. In these cases, octopamine injection is ineffective, suggesting that the stress-induced release of octopamine (50) is sufficient to saturate the SR response. Preexperimental stress leading to an octopaminergic modulation has similarly been discussed for various muscles (19, 40). A saturation of the SR response might also be reached during flight because octopamine injection causes no further increase in SR activity (J.-M. Ramirez, unpublished observation). Therefore, the natural function of octopamine probably is not so much to increase the SR activity but rather to induce the saturated response characteristics of the SR. As a consequence, wing movements cause very constant high discharge responses in the SR, an important prerequisite for the SR to exert its effects on the flight-rhythm generator. A similar modulatory effect might also occur in the other important wing proprioceptor, the tegula, which is also modulated by octopamine (66).

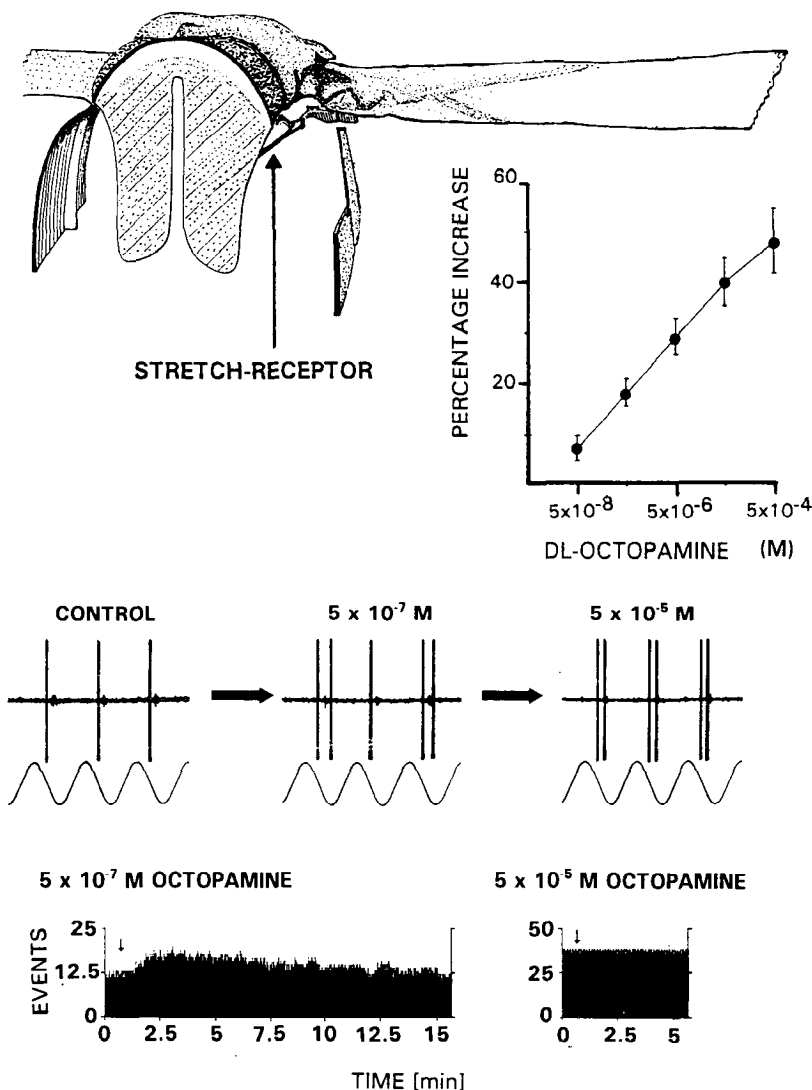


Figure 3 Octopaminergic modulation of the stretch receptor (SR). The SR is a single-celled sense organ located at the base of each wing. The responsiveness of this sense organ to passive sinusoidal wing movements (recordings: upper trace, extracellular recording from SR; lower trace, sinusoidal stimulus to passively move the wing) is enhanced in a dose-dependent manner (see dose-response curve) by octopamine injected into the hemolymph of intact locusts. SR activity reaches a peak 1.5 minutes after DL-octopamine injection and declines gradually over the next 14 min (left sequential-spike histogram). After octopamine has been injected over short intervals, even a high concentration of octopamine (5×10^{-5} M) does not further enhance the SR activity (right sequential-spike histogram). (Dose response curve, recordings, and spike histograms are from Ref. 65; drawing modified from Ref. 58.)

Modulation of primary afferents, as described for the octopaminergic modulation of the SR, is a general phenomenon found in other invertebrates. In stick insects, octopamine modulates the femoral chordotonal organ (J.-M. Ramirez, unpublished observations), and in crustaceans, octopamine, serotonin, and proctolin modulate various mechanoreceptors (see 54). In the mollusc *Aplysia*, a mechanosensory neuron is modulated by dopamine, acetylcholine, serotonin, and the peptide Phe-Met-Arg-Phe-amide (FMRFamide) (7). It is therefore hypothesized that modulatory substances in invertebrates might perform a function similar to the efferent neuronal control found in vertebrates. In both cases, the degree of modulation is variable and depends on the animal's activity state and behavior (60, 65).

OCTOPAMINERGIC MODULATION OF FLIGHT MUSCLES

In the previous sections, we discussed the role of octopamine in mobilizing lipids from the fat body stores, inducing active-membrane properties in flight interneurons, and enhancing the sensitivity of wing proprioceptors. Octopamine also affects flight muscles. On the one hand, it regulates metabolism of flight muscles, stimulating the oxidation of glucose, trehalose, butyrate, and diacylglycerol in working perfused thoracic muscles (9), rather like catecholamines stimulate oxidative metabolism in mammals. On the other hand, octopamine modulates the contraction kinetics of the tergocoxal and dorso-longitudinal (DL) muscles of locusts (9, 40, 88). The main modulatory effect of octopamine is an increase in the amplitude of twitch tension in the DL and tergocoxal muscles, which leads to an increased power output (40, 88). The twitch duration is also affected by octopamine. In the DL muscle at 20°C, twitch duration decreases in the presence of octopamine, but the duration is less influenced at 30°C (88). In the tergocoxal muscle at 30°C, however, the twitch duration increases slightly during octopamine perfusion (40). Significant modulatory effects of these contraction properties are obtained at octopamine concentrations comparable to those found in the hemolymph during the first minutes of flight. These octopaminergic effects have several consequences for flight behavior. The increased twitch amplitude leads to an increase in the amount of force generated by the flight muscles, which could be important during take off (40, 88). The shorter twitch duration observed for the DL muscle at low temperature allows a better synchronization in the contraction of the flight muscles at the onset of flight. At this time, body temperatures are still low, and the octopamine-induced decrease in twitch duration is maximal (88). Thus, the octopaminergic modulation of flight muscles seems to represent an energy-saving adaptation prior to warming up (88).

THE ROLE OF ENDOGENOUSLY RELEASED OCTOPAMINE IN THE CONTROL OF LOCUST FLIGHT

The functional significance of octopaminergic modulation for flight performance is usually assessed from data obtained *in vitro* or *in vivo* in the quiescent animal. This practice may lead to an underestimation of the physiological effect during flight. Dose-response curves, as were obtained for the SR, the flight muscles, and the adipokinetic action of octopamine, allow only a very rough estimation of the modulatory effects during flight, if concentrations of exogenously applied octopamine are compared with those occurring in the hemolymph. One difficulty is that most investigations have used DL-octopamine and not the naturally occurring active stereo-isomer, D-octopamine (24). However, a more significant problem is that local concentration gradients, dynamics of octopamine release, and interaction with other naturally released hormones have not been assessed. In order to understand the natural role of octopamine during flight, these important aspects of octopaminergic modulation have to be taken into consideration.

The three adipokinetic hormones, octopamine, AKH I, and AKH II, were treated in an earlier section as independent neurohormones. However, interactions may occur between two or more of these neurohormones, although this possibility has never been tested. For example, exposure of the fat body to octopamine may be necessary to prime the fat body to respond more efficiently to the peptidergic hormones. Indeed, *in vitro* fat body responds poorly to AKHs with regard to lipid release. The poor response has generally been attributed to problems of viability *in vitro* or a deficiency of hemolymph lipoprotein in the incubation medium. Perhaps, however, more subtle changes induced by octopamine occur *in vivo* but are lacking *in vitro*. Another possible interaction involves synergism between octopamine and AKHs. In *Rhodnius prolixus*, both serotonin and a peptidergic diuretic hormone (DH) have diuretic activity on Malpighian tubules and both are released following a blood meal (see 36). Whereas the time course of serotonin release is short term, like octopamine in locusts, that of DH release is longer (see 3). Barrett & Orchard (3) suggested that serotonin and DH may act synergistically on the Malpighian tubules, and indeed this has now been confirmed (reported in 39). A parallel situation may occur for locust fat body with octopamine and AKHs. This may well be a fruitful study to undertake.

Not only the interaction with other hormones but also local concentration gradients may play important roles in the modulatory processes occurring during flight. Under natural conditions, octopamine might be released in the direct vicinity of flight interneurons, muscles, or proprioceptors at concentrations that are quite different from the concentrations measured in the entire hemolymph or nerve cord. A local increase in octopamine concentration may

readily be decreased by the natural uptake/removal systems. These systems may be important in precisely regulating the modulatory effects. Despite the obvious importance of this aspect of modulation, we are just beginning to understand the mechanism of octopamine release during flight, and it may well be that different octopaminergic neurons fulfill different modulatory tasks during flight. Octopamine has been shown to be associated with neurohemal organs in the insect nervous system (18), including the corpora cardiaca and perisymphatic organs lying on the unpaired nervous system in each segment. Clearly, these are in an ideal position to quickly flood the hemolymph with octopamine, an important prerequisite for adipokinetic function.

More important for the modulation of flight muscles and wing proprioceptors might be the local release of octopamine by the DUM neurons. Some of these neurons end blindly as neurosecretory terminals on various muscles and therefore in the vicinity of proprioceptors (30). Intracellular recordings obtained in deafferented and intact flying locusts reveal that the thoracic DUM cells of the morphological type DUMDL, DUM3, DUM3,4, and DUM5 are indeed activated at the onset of flight (65) (Figure 4). The contribution of spill-over from these DUM cells to the modulation of the SR has been demonstrated by contralateral stimulation of the nerves 1, 3, and 5, which contain axons of the bilaterally projecting DUM cells. Contralateral stimulation of DUM axons in either one of these nerves results in an enhancement of the SR response similar to that seen following octopamine application, indicating that spill-over from DUM cells may contribute to the octopaminergic modulation of the SR (65) (Figure 4). In addition, the dorso-longitudinal muscles are modulated by DUM neuron activation (88). Maximal modulation is reached when stimulating the DUMDL at frequencies between 5–10 Hz. This frequency range correlates with the spike frequencies occurring during flight (65). Therefore, modulation of the SR and the flight muscles during flight probably mainly results from local release of octopamine from DUM cells. The modulation by octopamine circulating in the hemolymph might therefore be regarded as a secondary effect (see 88).

The question arises as to whether the activation of DUM cells is also responsible for a central modulation of flight interneurons (71, 72) and for the increase in octopamine concentration in the metathoracic ganglion during flight (16). It is known that under normal conditions at least nine DUM cells in each ganglion are simultaneously activated at the onset of flight (65). Electron-microscopical studies have shown that these DUM cells make output synapses within the thoracic ganglia (87), in neuropile areas overlapping with those of flight interneurons (67). Although these findings suggest that DUM cells contribute to the modulation of central neurons, the direct evidence is still missing. For example, it has not been possible to simulate the oc-

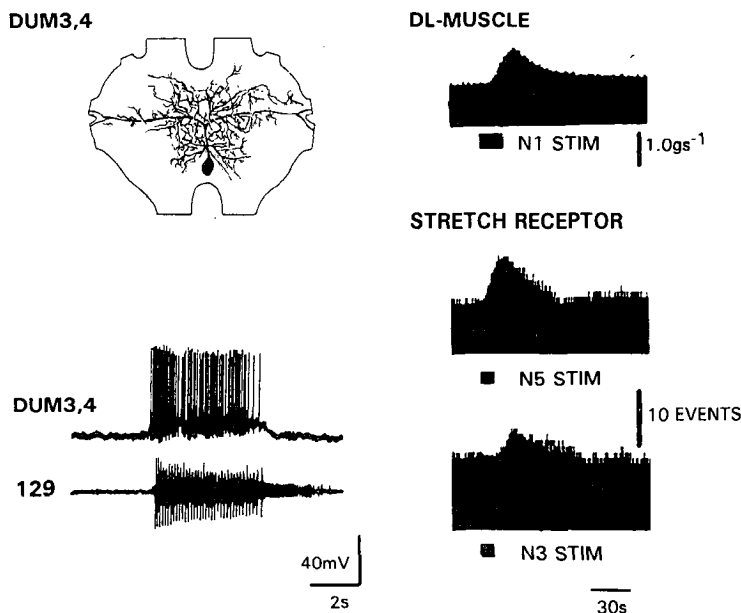


Figure 4 The role of dorsal unpaired median (DUM) neurons in modulating DL muscle and stretch receptor. At least nine DUM cells in each thoracic ganglion project bilaterally into the peripheral nerves 1, 3, 4, and 5 (drawing of one DUM3,4 neuron projecting into the thoracic nerves 3 and 4). All DUM neurons projecting into these nerves were activated during flight (recording: upper trace, intracellular recording from DUM3,4 neuron; lower trace, electromyograph recording from the wing depressor muscle 129). Octopamine released from DUM cells modulates DL muscle and SR. Contralateral stimulation of nerve 1 (30 Hz), presumably stimulating DUMDL, enhances the relaxation rate of the DL muscle. The responsiveness of the SR is enhanced following contralateral stimulation of thoracic nerves 5 and 3 (10 Hz), stimulating presumably DUM cells projecting in these nerves. (Sequential histogram from DL muscle modified from Ref. 88; other histograms, the recording, and drawing are from Ref. 65.)

topaminergic modulation of flight interneurons by intracellular stimulation of single DUM cells (J.-M. Ramirez, unpublished observation). This means that either the activation of more than one DUM cell is necessary to induce bursting properties in flight interneurons or that DUM cells play only a secondary role in central modulation of flight interneurons. Thus, central octopaminergic modulation may result from octopaminergic interneurons. Immunohistochemical studies have shown that several intersegmental interneurons stain with an antibody against octopamine (83), but further experiments are needed to determine whether these neurons are responsible for the central modulatory effects. Finding the link between modulatory neurons and the induction of bursting properties, as has already been achieved for the crustacean stomato-

gastric system (for a review see 32), will be a task for the future and will certainly give new insights into the functional role of octopaminergic modulation.

OCTOPAMINERGIC MODULATION OF INSECT BEHAVIOR

All data presented demonstrate that octopamine plays an important role in the regulation of locust flight. The events occurring at the onset of flight are summarized in Figure 5 and can be described as follows: At the onset of flight DUM neurons, probably representing only one population of octopamine neurons, are activated by wind stimuli via as yet unidentified wind sensory interneurons. Wind-sensory information, presumably via flight-initiating interneurons, leads, after a latency of approximately 70 ms, to the activation of the neuronal circuit for flight and, via motoneurons, to the initiation of flight. However at this time, all of the thoracic DUM cells (as well as other octopaminergic neurons) are already activated, releasing octopamine peripherally into the hemolymph and into the direct vicinity of flight muscles and proprioceptors. This peripheral release leads to an increase in responsiveness of wing proprioceptors as well as to an increase in the power output and relaxation rate of flight muscles. DUM cells, and perhaps other octopaminergic neurons, also release octopamine centrally into the dorsal flight neuropile, which leads to the induction of bursting properties in flight interneurons. Activity in DUM cells is maintained via flight-coupled excitatory feedback from as yet unknown sources.

Although increased octopamine levels at this early phase of flight lead to the first release of lipid from the fat body, the major source of metabolic energy for muscular activity is still trehalose, and hemolymph trehalose levels drop precipitously during the first few minutes of flight (4). This drop in hemolymph trehalose may be a trigger that in turn activates, via unknown neurons, a population of octopaminergic neurons that send axons along NCC II to synapse with the neurosecretory cells in the glandular lobe of the corpus cardiacum, thereby stimulating the release of AKH I and II. The release of AKHs at this later phase of flight leads to a further increase in hemolymph lipid, which now becomes the major energy source to fuel muscle activity. Thus, octopamine, acting as a neuromodulator, neurohormone, and neurotransmitter, coordinates and biases metabolic, muscular, and neuronal events associated with flight.

A review dealing with the octopaminergic modulation of locust flight might lead to the impression that one substance is specialized for only one behavior. This is certainly not the case since octopamine is not the only substance affecting locust flight, and locust flight is not the only behavior affected by

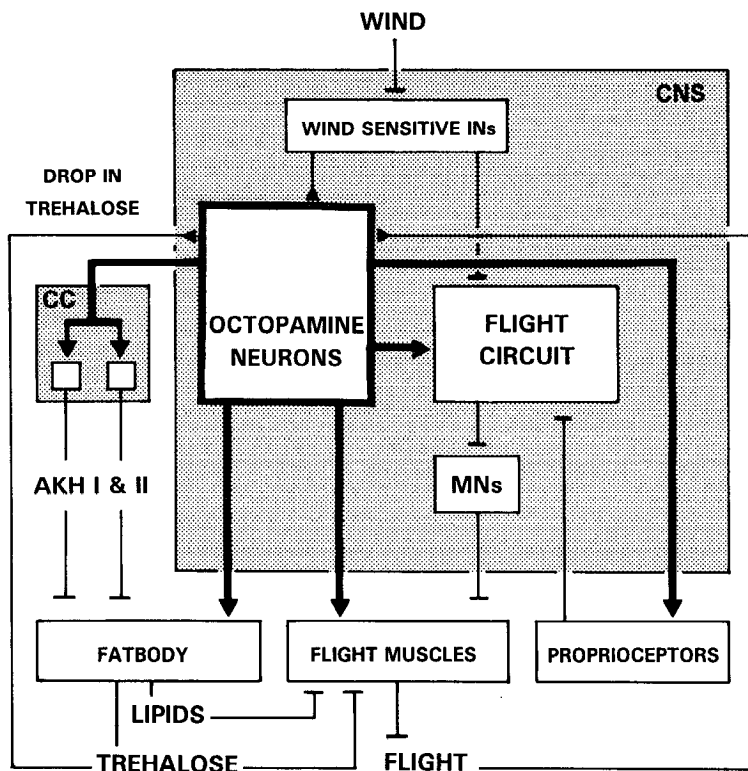


Figure 5 Events associated with octopamine during locust flight. For explanation see text.

octopamine. Thus, for a more general understanding of how behavior is regulated by modulatory substances, one must consider other modulatory influences as well.

As was discussed for the regulation of hemolymph lipids, octopamine plays a major role only during the first minutes of flight. Later in flight, lipid regulation appears to be controlled by the peptide hormones AKH I and II. This alteration in lipid regulation is reflected in different time courses of the hemolymph titers (42). A similar complex cascade of substances might also be important for the control of all other described neuronal and muscular events. Although this possibility has so far not been investigated in detail, the SR and the flight rhythm generator are also affected by other amines. The SR is, for example, as sensitive to serotonin as it is to octopamine, while dopamine is ineffective (65). On the other hand, flight rhythmic activity is effectively suppressed by serotonin (80) while dopamine has as much potential as octopamine for inducing flight rhythmic activity in the locust and the moth

Manduca sexta (12, 80). However, the functional role of serotonin and dopamine in the control of flight is unknown, because there is no clear evidence that these amines are released into the hemolymph or into the CNS during flight. Thus, further investigations will be necessary to examine whether the neuronal events occurring during flight are modulated by a cascade of different substances, as has been demonstrated for lipid regulation.

As mentioned above, octopamine is not an amine specialized only for flight. It is released during flight (24) but also during stress (41, 50) and general locomotor activity (15). This amine has therefore often been discussed as the functional equivalent of the vertebrate fight-or-flight hormone, noradrenaline (28, 41). Based upon investigations on leg muscles, Evans & Siegler (20) have proposed that octopamine levels in the hemolymph are elevated as part of a general peripheral arousal mechanism that alters the responsiveness of insect muscles from a postural to a dynamic mode. This is certainly not only valid for the leg muscles investigated, but also for the flight muscles (40, 88). More recent findings suggest that this function of octopamine proposed for muscles can be extended to a more general arousal function including the peripheral and central nervous system. In the peripheral nervous system, proprioceptors play important roles in the control of postural and locomotor behaviors. Octopamine enhances the responsiveness to sensory stimuli not only in wing proprioceptors but also in leg proprioceptors, such as the femoral chordotonal organ (J.-M. Ramirez, unpublished observation). The transformation from a passive into a dynamic mode is also the principle for the neuronal elements within the CNS. As has already been mentioned, the membrane properties of flight interneurons are altered in a way that leads to a transformation of the neuronal circuit for flight from a passive into a dynamic mode. A similar central effect is known for leg reflexes, which in general are highly affected by the behavioral state of the animal (60). Octopamine induces, in stick insects, the transformation of leg reflexes from a postural into a dynamic mode by suppressing the resistance reflex, a typical postural reflex (8), and simultaneously enhancing the probability of "active reaction," a reflex that is typical during locomotion. This is consistent with findings in locusts in which octopamine induces the activation of leg movements (78). A behavior that is also highly modulated by the state of the animal is the escape response to cercal wind stimulation, which is best described for the cockroach. In this insect, octopamine enhances the responsiveness to wind stimulation (23). An enhancement in synaptic transmission could, at least in part, produce this result (10). Similarly, respiration is a very state-dependent behavior that is enhanced during stress (21). For example, ventilation in the dobsonfly is increased following octopamine injection (5). Also consistent with its general arousal function is the finding that octopamine inhibits oviposition digging and oviduct contractions in female locusts (35, 78). Female locusts are

particularly vulnerable during digging and egg laying, and it is therefore advantageous to suppress these behaviors during times of stress.

All these findings are consistent with the notion that octopamine is part of a general arousal mechanism that transforms the insect from a quiescent into a dynamic state. Such an alteration in the state of the insect parallels very closely the situation in higher vertebrates. The state of vertebrates is altered in anticipation of movements as well as during stress (60). This alteration of the vertebrate into a new state ("set") (60) includes an increase in muscle tone, alteration in various reflex parameters (e.g. decrease in gain and phase reversals), alteration in proprioception (e.g. spindle activity), and cutaneous and proprioceptive transmission. These changes are influenced, though not exclusively controlled, by neuronal interactions. The data presented for insects suggest that control is mainly achieved by the amine octopamine. However, to serve a similar anticipatory function, octopamine must be released in anticipation of movements. That this is indeed the case is indicated by the finding that the octopaminergic DUM cells are activated prior to flight activity (65), prior to any walking, jumping, or kicking movement, and whenever locusts become agitated or disturbed (31). The interesting question arises as to whether a specific insect behavior (e.g. flight, walking, or kicking) can be biased in a controlled and specific manner towards an active state by the activation of certain DUM cells. According to the orchestration hypothesis, this could be done by activating a specific population [a specific subcluster (77)] of DUM cells that initiate, in concert, a specific behavior. This hypothesis is supported by the finding that injection of octopamine can initiate a specific behavior depending on the particular neuropile region of injection and that stimulation of single DUM cells does not elicit any behavior (77). Similarly, in bees, different aspects of olfactory learning are modulated depending on the site of injection (see 6). The finding that specific behaviors can be elicited in specific areas does not, however, indicate that octopamine, under natural conditions, is also released in a specific manner. In fact, recordings from thoracic DUM cells indicate that the DUM cells are activated at the onset of flight, walking, kicking, or agitation (31, 65), favoring a more general arousal function than control of a specific behavior. A general arousal function is also supported by the measurements that indicate that octopamine levels increase unspecifically during stress and flight (15, 50). General arousal effects are also found for bee olfactory learning (6). Thus, both general and specific modulatory effects might be involved in the control of insect behavior. However, the question as to whether octopamine acts specifically and/or generally remains speculative as long as the source of modulation has not been identified for each aspect of behavior. This task will certainly be accomplished in the near future. Over the past decade, the flight system of the locust has proven to be an ideal system for study of neuromodulation from

the behavioral to the cellular level, and we can expect new, exciting information to be forthcoming on how animal behavior is regulated by the interaction of modulatory substances with various muscular, metabolic, and neuronal events.

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